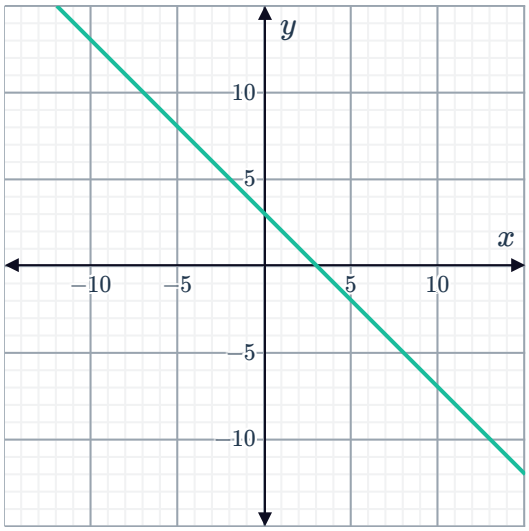
Constant rates of change

1 State whether the following functions have a constant or variable rate of change:

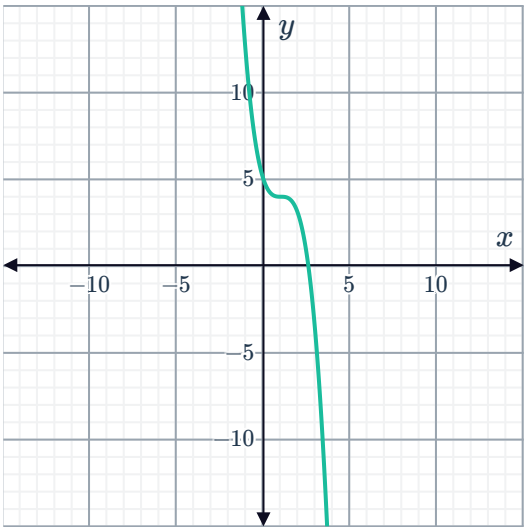
a $2x + y + 3 = 0$

b $y = 2^x$

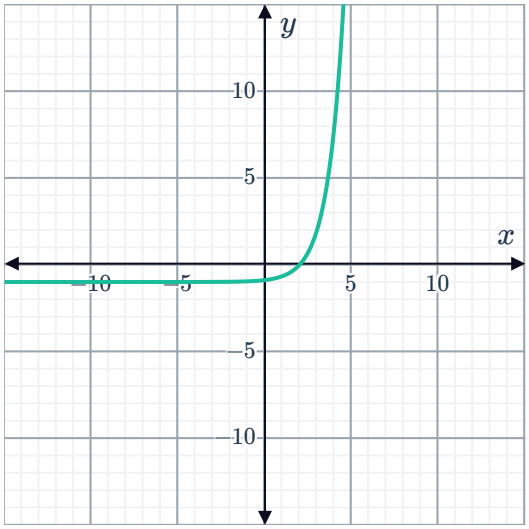
c



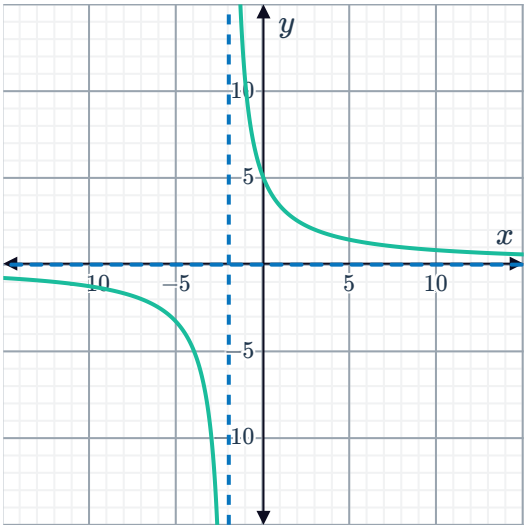
d



e



f



2 Does the function that passes through the following points:

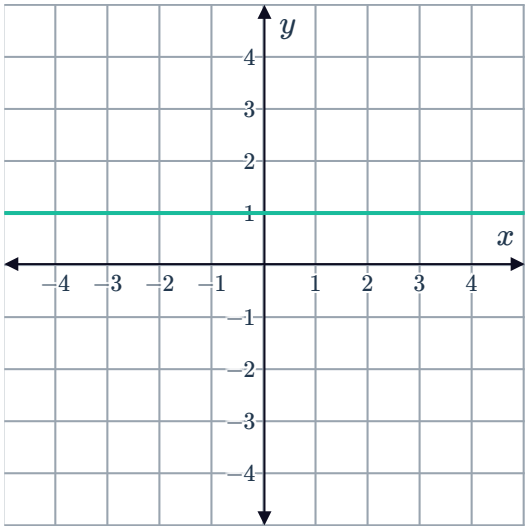
$$\{(-2, -5), (1, -20), (2, -25), (7, -50), (9, -60)\}$$

have a constant or a variable rate of change?

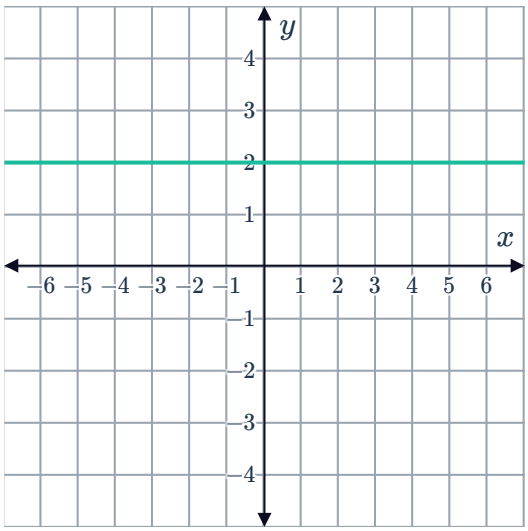
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Instantaneous rates of change

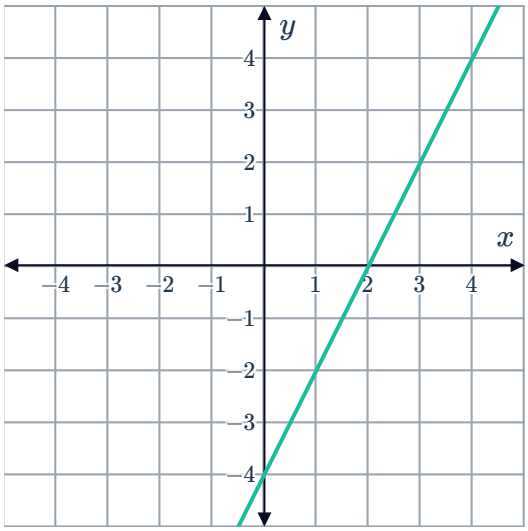
3 Consider the graph of the function $f(x) = 1$. If we were to draw a tangent to the function for any x-value, find the gradient of that tangent.



4 Consider the graph of the function $f(x) = 2$: Find the gradient of the tangent at the point $x = 3$.



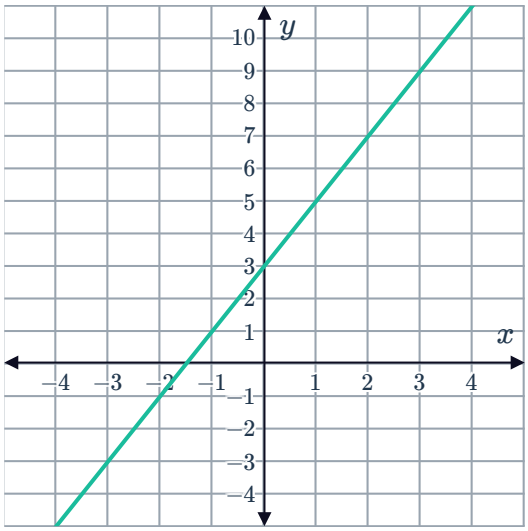
5 Consider the graph of the function $f(x) = 2x - 4$. If we were to draw a tangent to the function for any x-value, find the gradient of that tangent.



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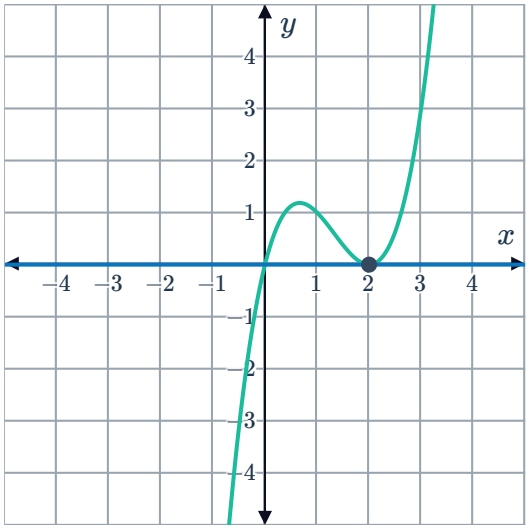
6 Consider the graph of the function $f(x) = 2x + 3$:

Find the gradient of the tangent at the point $x = -2$.

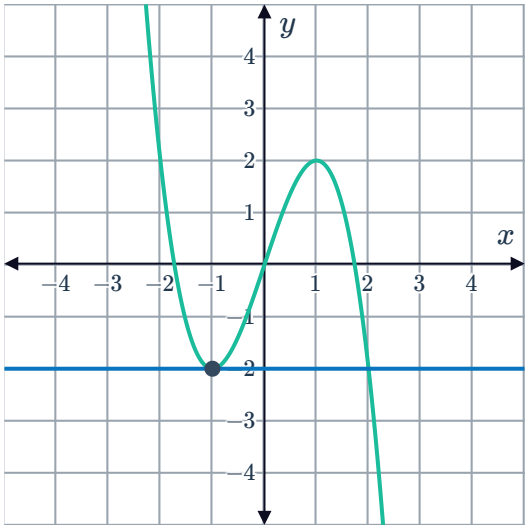


7 For each of the following graphs, find the gradient of the tangent at the given point, and hence the instantaneous rate of change:

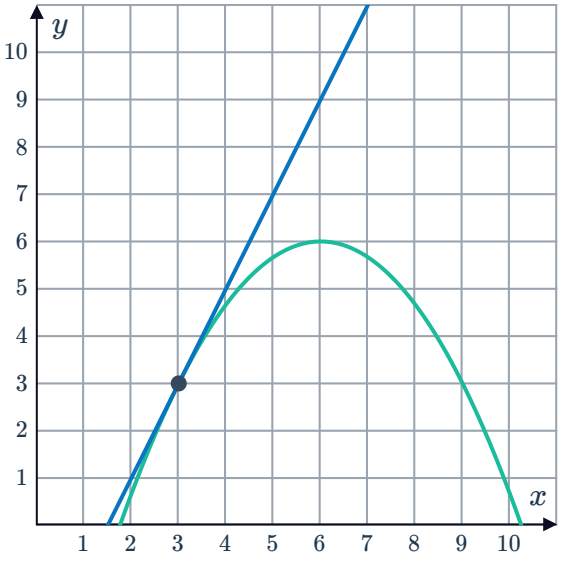
a



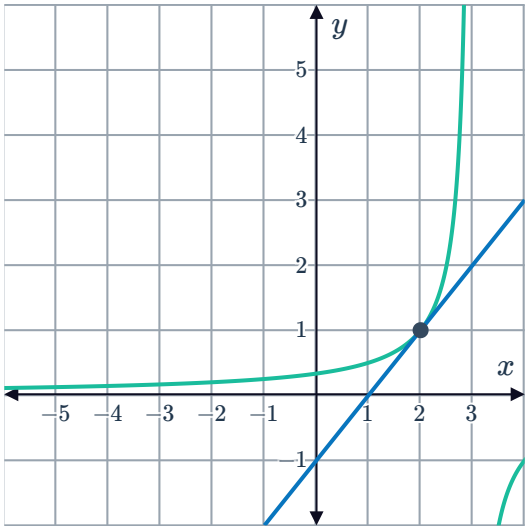
b



c

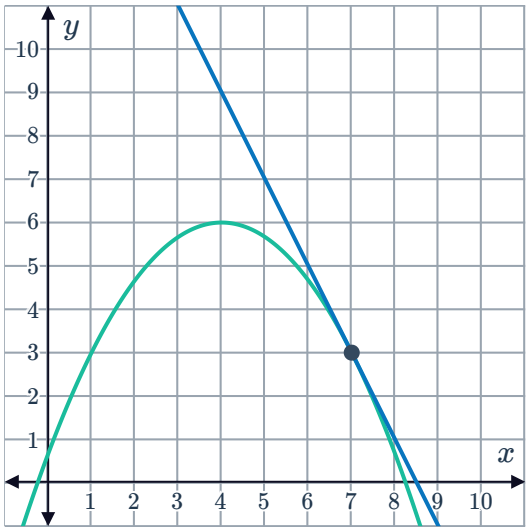


d

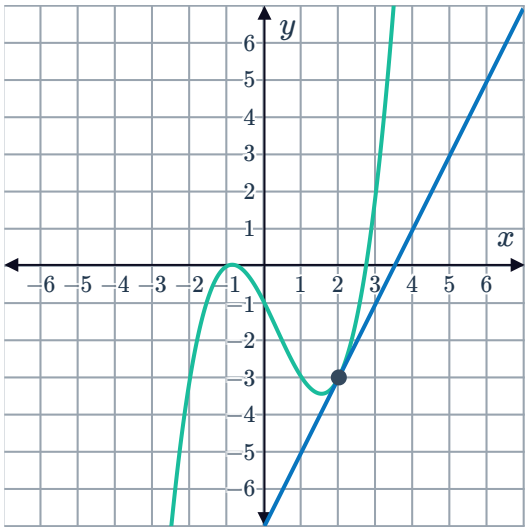


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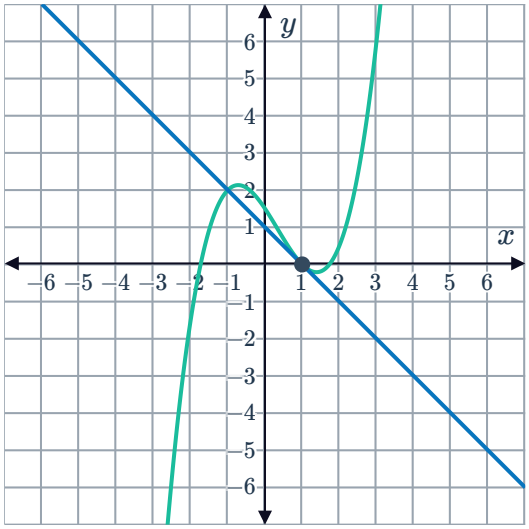
e



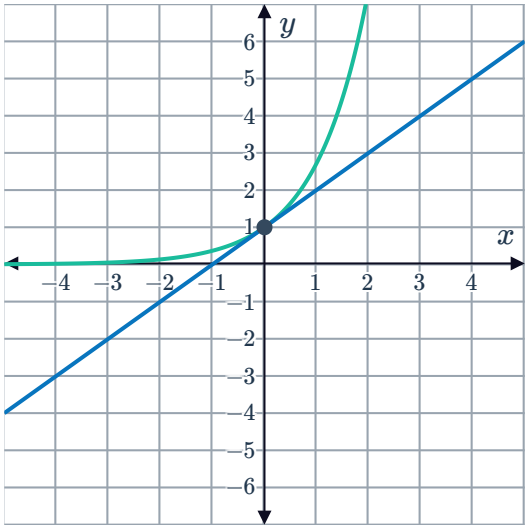
f



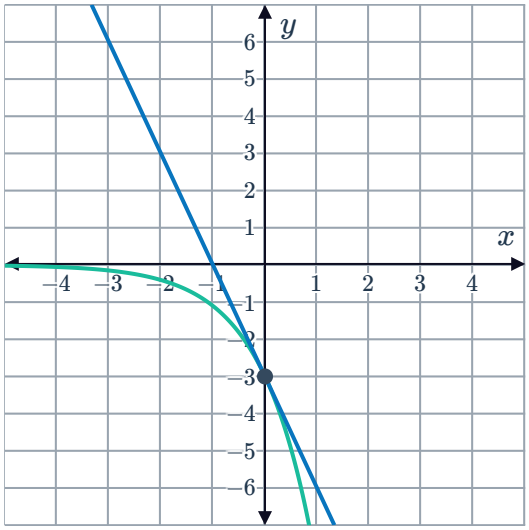
g



h



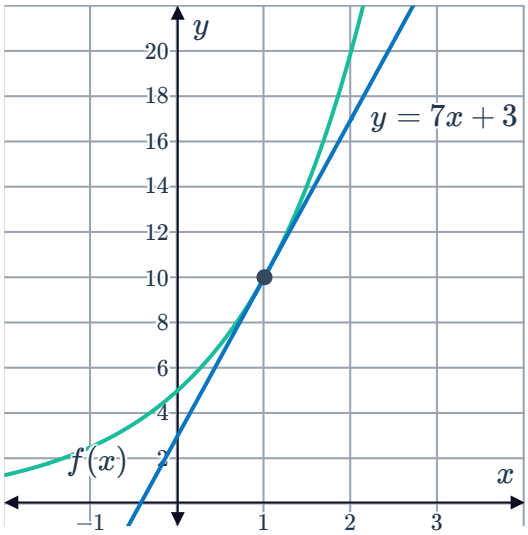
i



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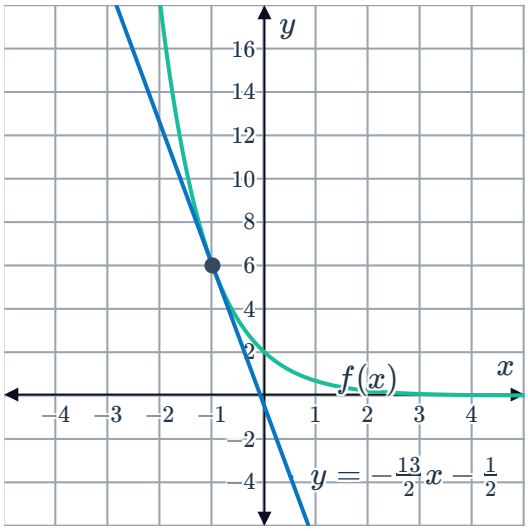
8 Consider the function $f(x) = 5(2)^x$ and the tangent line given by the equation $y = 7x + 3$ at $x = 1$:

Calculate the instantaneous rate of change at $x = 1$.



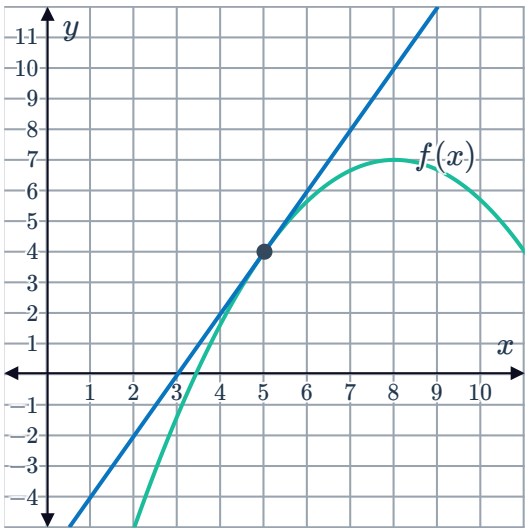
9 Consider the function $f(x) = 2(3)^{-x}$ and the tangent line given by the equation $y = -\frac{13}{2}x - \frac{1}{2}$ at $x = -1$:

Calculate the instantaneous rate of change at $x = -1$.



10 Consider the function $f(x) = -\frac{(x-8)^2}{3} + 7$ and a tangent line at $x = 5$:

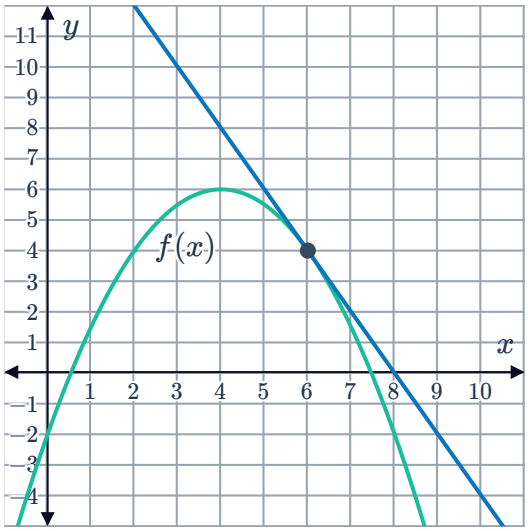
Calculate the instantaneous rate of change at $x = 5$.



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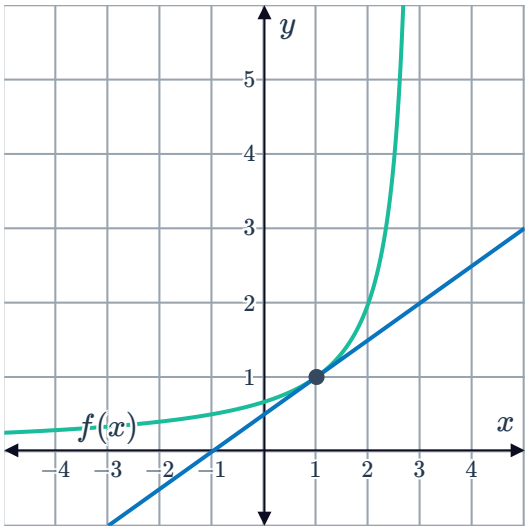
11 Consider the function $f(x) = -\frac{(x-4)^2}{2} + 6$ and a tangent line at $x = 6$:

Calculate the instantaneous rate of change at $x = 6$.



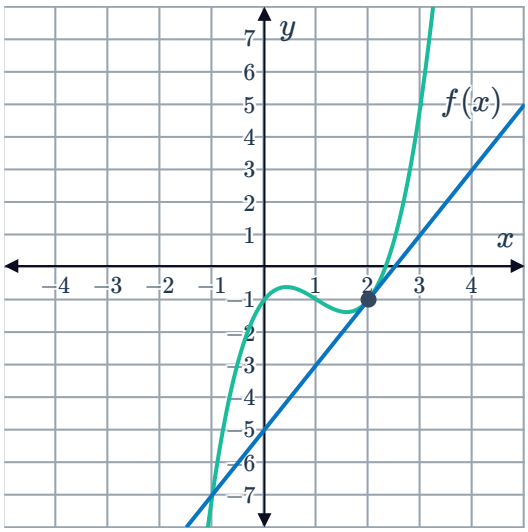
12 Consider the function $f(x) = -\frac{2}{x-3}$ and a tangent line at $x = 1$:

Calculate the instantaneous rate of change at $x = 1$.



13 Consider the function $f(x) = x^3 - 3x^2 + 2x - 1$ and a tangent line at $x = 2$:

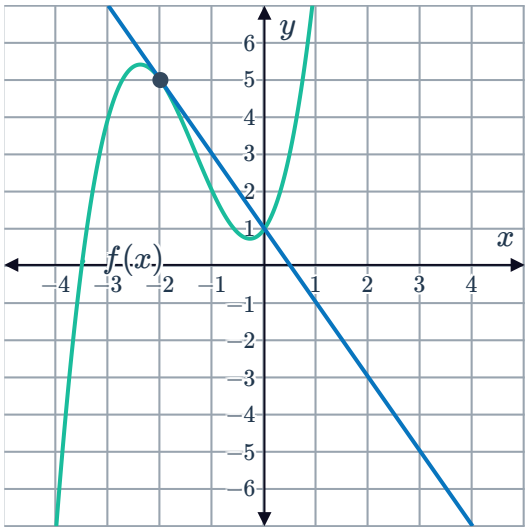
Calculate the instantaneous rate of change at $x = 2$.



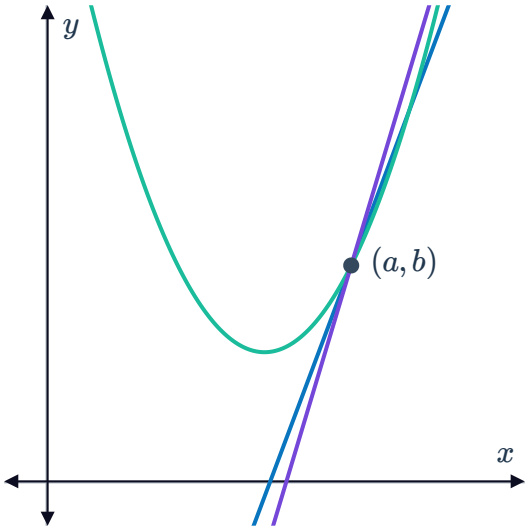
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14 Consider the function $f(x) = x^3 + 4x^2 + 2x + 1$ and a tangent line at $x = -2$:

Calculate the instantaneous rate of change at $x = -2$.



15 Consider the following figure. One secant line through the given point has the equation $y = 2.02x - 9.14$, while another secant line through the same point has the equation $y = 2.01x - 9.07$:



a Find the value of a.

b Find the value of b.

c Find the gradient of the tangent at this point to the nearest integer.

16 Consider the function $f(x) = x^3$.

a By filling in the table of values, complete the limiting chord process for $f(x) = x^3$ at the point $x = 1$. Round your answers to four decimal places if needed.

b Find the instantaneous rate of change of $f(x)$ at $x = 1$.

a	b	$h = b - a$	$\frac{f(b) - f(a)}{b - a}$
1	2	1	
1	1.5		
1	1.1		
1	1.05		
1	1.01		
1	1.001		
1	1.0001		

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17 Consider the function $f(x) = 2x^2$.

a By filling in the table of values, complete the limiting chord process for $f(x) = 2x^2$ at the point $x = 1$.

b Find the instantaneous rate of change of $f(x)$ at $x = 1$.

a	b	$h = b - a$	$\frac{f(b) - f(a)}{b - a}$
1	2	1	
1	1.5		
1	1.1		
1	1.05		
1	1.01		
1	1.001		
1	1.0001		

18 Consider the function $f(x) = 4^x$.

a By filling in the table of values, complete the limiting chord process for $f(x) = 4^x$ at the point $x = 0$. Round your answers to four decimal places if needed.

b Find the instantaneous rate of change of $f(x)$ at $x = 0$ correct to three decimal places.

a	b	$h = b - a$	$\frac{f(b) - f(a)}{b - a}$
0	1	1	
0	0.5		
0	0.1		
0	0.05		
0	0.01		
0	0.001		
0	0.0001		

19 Consider the function $f(x) = -x^2 + 5$.

a By filling in the table of values, complete the limiting chord process for $f(x) = -x^2 + 5$ at the point $x = 1$.

b Find the instantaneous rate of change of $f(x)$ at $x = 1$.

a	b	$h = b - a$	$\frac{f(b) - f(a)}{b - a}$
1	2	1	
1	1.5		
1	1.1		
1	1.05		
1	1.01		
1	1.001		
1	1.0001		

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20 Consider the function $f(x) = \dfrac{5}{x}$.

a By filling in the table of values, complete the limiting chord process for $f(x) = \dfrac{5}{x}$ at the point $x = 1$. Round your answers to four decimal places if needed.

b Find the instantaneous rate of change of $f(x)$ at $x = 1$.

a	b	$h = b - a$	$\dfrac{f(b) - f(a)}{b - a}$
1	2	1	
1	1.5		
1	1.1		
1	1.05		
1	1.01		
1	1.001		
1	1.0001		

Applications

21 The table shows the linear relationship between the temperature ($^\circ F$) on a particular day and the net profit, in dollars of a store. Find the rate of change of the net profit.

Temperature	-4	-3	-1	3
Net profit	49	39	19	-21

22 Due to adverse market conditions, Tobias and Gwen have had to reduce the number of staff at their respective companies, which currently have 310 staff.

Tobias plans on reducing staff numbers by 41 each year for the next five years, while Gwen plans on reducing staff numbers by 8 next year, 16 in the year after next, 24 in the year after that, and so on.

Year	0	1	2	3	4	5
Tobia's staff						
Gwen's staff						

a Complete the table. Note that year 0 represents the current year.

b What will be the rate of change in the size of Tobias's staff between years 3 and 4?

c What will be the rate of change in the size of Gwen's staff between years 3 and 4?

d Is the function representing the size of Tobias's staff linear or non-linear? Explain your answer.

e Is the function representing the size of Gwen's staff linear or non-linear? Explain your answer.

f Whose staff will decrease more quickly between years 3 and 4?

g Whose staff will decrease more quickly between years 10 and 11?

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23 Mario and Christa have both recently started their own small businesses, and initially neither of them had any clients. In it's first year, Mario's client base increased by 27 per month, while Christa's client base increased by 6 in its first month, 12 in its second month, 18 in its third month, and so on.

a Complete the following table:

Month	0	1	2	3	4	5
Mario's clients						
Christa's clients						

b Is the function representing the size of Mario's client base linear or non-linear?

c Is the function representing the size of Christa's client base linear or non-linear?

d Find the rate of change in the size of Mario's client base between months 2 and 3.

e Find the rate of change in the size of Christa's client base between months 2 and 3.

f Whose client base increased more quickly between months 2 and 3?

f Whose client base increased more quickly between months 4 and 5?

24 Harry and Skye each have $\$2000$ in savings to invest. Harry chooses to put his savings in a fund that generates a return of $\$60$ each month, while Skye chooses to put her savings in a fund that generates a return of 2% each month.

a Complete the following table:

Month	0	1	2	3	4	5
Value of Harry's investment						
Value of Skye's investment						

b Find the rate of change of Harry's investment between months 2 and 3.

c Find the rate of change of Skye's investment between months 2 and 3.

d Is the function representing the value of Harry's investment linear or non-linear?

e Is the function representing the value of Skye's investment linear or non-linear?

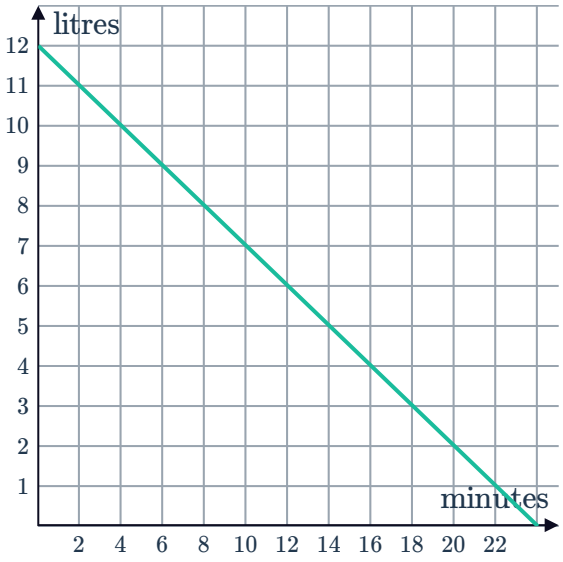
f Whose investment increased the most in value between months 2 and 3?

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25 A bucket containing water has a hole through which the water leaks. The graph shows the amount of water remaining in the bucket after a certain number of minutes:

a Find the gradient of the line.

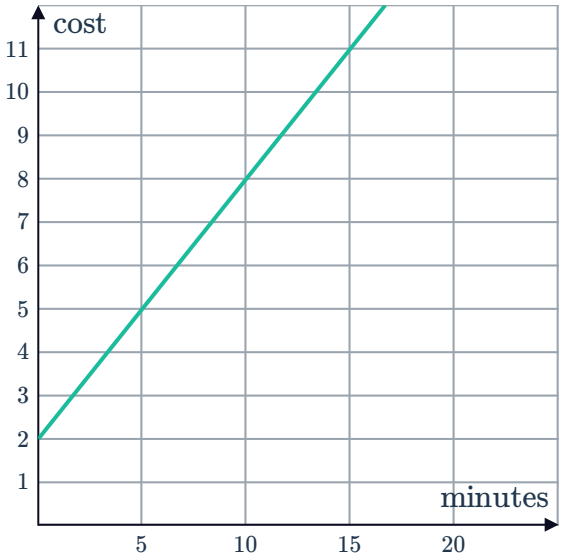
b Describe the meaning of the gradient in context.



26 The graph shows the cost, in dollars, of a phone call for different call durations in minutes:

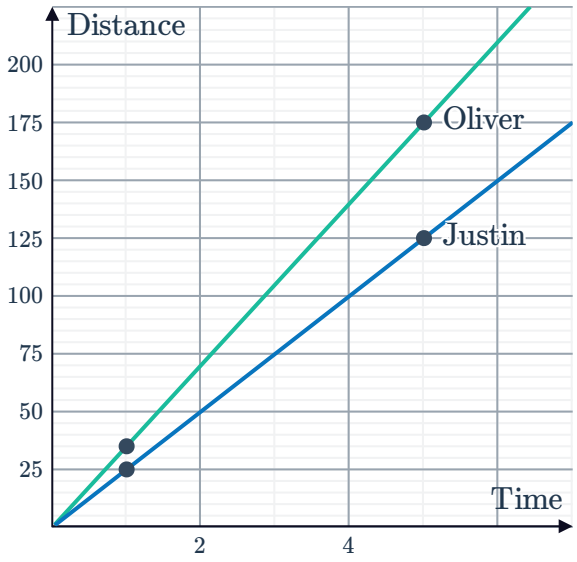
a Find the gradien of the line.

b Describe the meaning of the gradient in context.



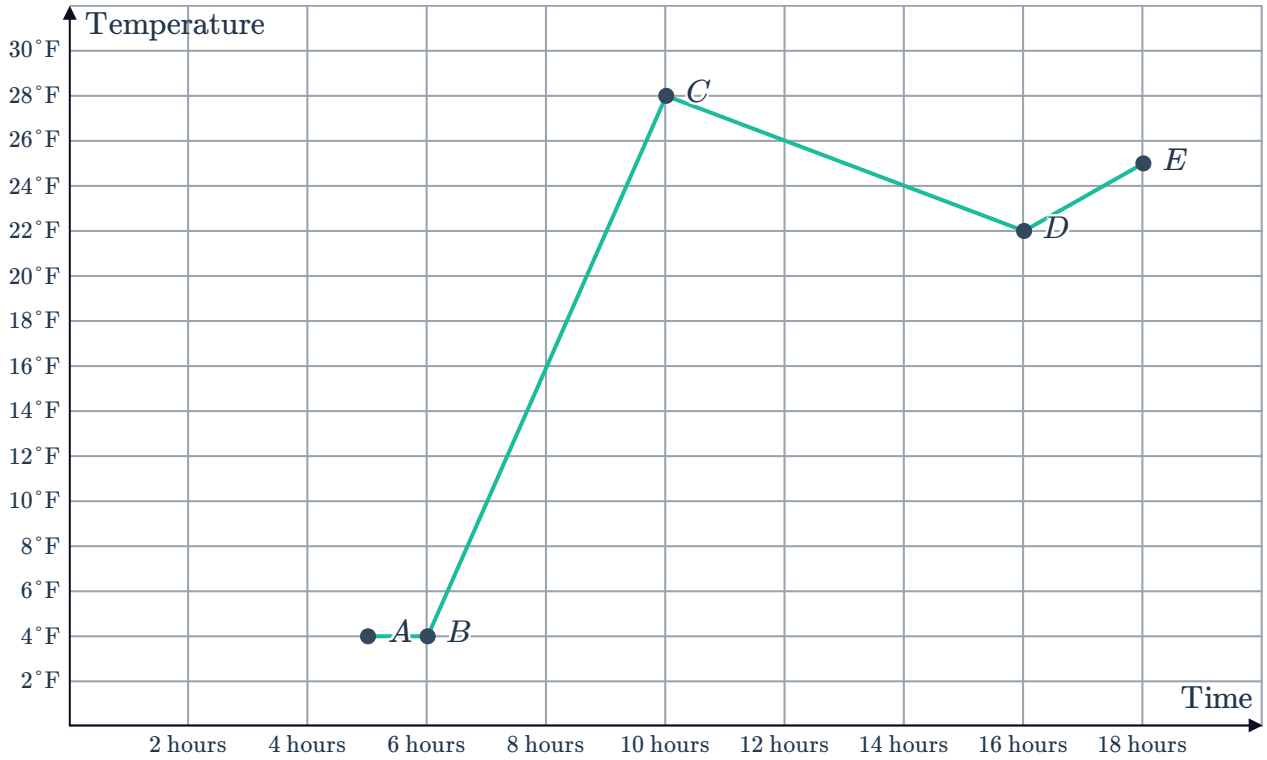
27 The graph shows the progress of two competitors in a cycling race.

Who is travelling faster and how much faster is he travelling?



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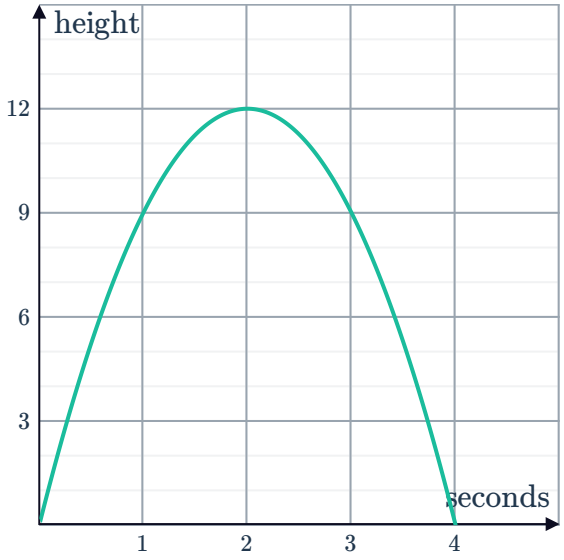
28 The graph shows the recorded temperatures at various hours after midnight in Antarctica:



Find the rate of change in the temperature between points:

a A and B **b** B and C **c** C and D **d** D and E

29 The graph shows the height of a cricket ball in metres after it is thrown.



a Find the rate of change of the height of the ball in the interval between:

- **i** When it is thrown and $t = 1$.
- **ii** $t = 1$ and when it is at its highest point.
- **iii** When it is at its highest point and $t = 3$.
- **iv** $t = 3$ and when it returns to the ground.

b In which of the above intervals is the ball travelling at its fastest speed?

f What do the negative rates of change in the interval between $t = 2$ and $t = 3$ and in the interval between $t = 3$ and $t = 4$ indicate?

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